Wholesale Cutlery Distribution Platform – Project Requirements Document

1. Introduction

1.1. Purpose

This document defines the requirements, architecture, and scope for developing a scalable web application for a wholesale cutlery distributor. The platform is intended to showcase business details, allow wholesalers to sign up (via invite or request), and provide robust management tools for admins to oversee orders, pricing, and product inventory.

1.2. Scope

The project comprises three major components:

- **Landing Page:** Public-facing site detailing the business, product highlights, and a request form for potential wholesalers.
- **Wholesaler Portal:** An invite-based, secure area where approved wholesalers can manage bulk orders, view pricing, access invoices, track payments, and analyze order insights.
- **Admin Portal:** A management dashboard for handling wholesaler requests, orders, invoices, pricing, product categories, and generating sales insights.

1.3. Stakeholders

- **Business Owners/Management:** Oversee the platform's functionality and business operations.
- **Admins:** Manage wholesalers, product inventory, orders, pricing, and customer queries.
- **Wholesalers:** Registered users who can request bulk orders, view flat price quotes, download invoices, and track payments.
- **Developers & Designers:** Build and maintain the platform.
- **Third-Party Service Providers:** HubSpot (for CRM), payment gateways, and logistics services.

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2. Project Objectives

- **Enhance Business Visibility:** Provide a professional landing page that communicates brand identity and product offerings.
- **Streamline Wholesale Onboarding:** Implement an invite/request mechanism to control wholesaler access.
- **Efficient Order Management:** Develop intuitive interfaces for wholesalers to place orders and track their statuses.
- **Robust Admin Controls:** Create a comprehensive dashboard to manage orders, invoicing, pricing, product categories, and overall sales insights.
- **Integration with HubSpot:** Seamlessly synchronize data with HubSpot to support internal CRM operations.

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3. Functional Requirements

3.1. Landing Page (Public Interface)

- **Business Overview:** Detailed company history, mission, product categories, and testimonials.
- **Product Showcases:** Highlight featured products and categories.
- **General Inquiry Form:** Allow visitors to ask questions or request more information.
- **Wholesaler Request Form:**
    - Users submit a “Become a Wholesaler” request.
    - Requests are routed to the Admin Portal for approval.

3.2. Wholesaler Portal

3.2.1. Registration & Access Control

- **Invite-Only Registration:**
    - Admin can invite potential wholesalers.
    - Alternatively, interested users submit a request which is then approved or rejected by the admin.
- **Authentication:** Secure login with JWT-based authentication.
- **Profile Management:** Update contact details and view account information.

3.2.2. Order Management & Dashboard

- **Bulk Order Request Interface:**
    - Wholesalers can select products, specify quantities, and submit orders.
    - Option to request custom quotes.
- **Order Tracking:**
    - Real-time order status (Pending, Approved, Shipped, Delivered).
    - Access to historical order data and analytics.

- **Pricing & Tier Details:**
    - Display flat price quotes and tier-based discounts.
    - Insight into tier benefits and eligibility.
- **Invoice & Payment Management:**
    - Download invoices.
    - View payment history and upload payment proofs.

3.3. Admin Portal

3.3.1. Wholesaler Management

- **Request Handling:**
    - Review and approve/reject wholesaler requests.
    - Invite wholesalers directly.
- **Profile & Tier Management:**
    - Assign pricing tiers.
    - Update wholesaler contact and business details.

3.3.2. Order and Invoice Management

- **Order Overview:**
    - View all incoming bulk orders with detailed status.
    - Approve orders and trigger invoice generation.
- **Invoice and Payment Processing:**
    - Manage invoice creation.
    - Update payment statuses as proofs are submitted.
- **Shipment Tracking:**
    - Integrate with logistics APIs to update tracking information.

3.3.3. Product & Pricing Management

- **Product Catalog Management:**
    - Add, edit, and delete products.
    - Organize products into categories and subcategories.
    - Set and update inventory levels.
- **Pricing Controls:**
    - Manage flat pricing and tier-based discount structures.
    - Configure discounts and promotional offers based on bulk order volumes.

3.3.4. Reporting & Insights

- **Sales Insights:**
    - Dashboard with key performance metrics such as total sales, order trends, and best-selling products.
- **Wholesaler Analytics:**

    - View order frequency, payment statuses, and tier performance.
- **Inventory Insights:**
    - Monitor stock levels and product popularity.

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4. Non-Functional Requirements

- **Scalability:**
    - Utilize a microservices-friendly architecture ensuring horizontal scalability.
- **Performance:**
    - Fast load times and efficient data processing using optimized queries and caching.
- **Security:**
    - Implement robust authentication (JWT) and role-based access control (RBAC).
    - Data encryption both in transit and at rest.
- **Reliability & Availability:**
    - Ensure 99.9% uptime through cloud-based deployment and auto-scaling.
- **Maintainability:**
    - Modular code structure, comprehensive documentation, and unit/integration testing.
- **Usability:**
    - Responsive and intuitive UI/UX for both wholesalers and admins.
- **Integration:**
    - Seamless HubSpot integration for CRM functionalities.
    - Support for payment gateways and logistics service providers.

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5. System Architecture & Tech Stack

5.1. Frontend

- **Framework:** React.js (Next.js preferred for server-side rendering and SEO benefits)
- **Language:** TypeScript
- **Styling:** Tailwind CSS or Material UI for component-based design
- **State Management:** React Query / SWR for efficient data fetching and caching

5.2. Backend

- **Runtime:** Node.js
- **Framework:** Express.js or NestJS (for a more structured and scalable approach)
- **Language:** TypeScript

- **API Type:** RESTful API or GraphQL for flexible data retrieval
- **Authentication:** JWT-based auth and role-based access control (RBAC)

5.3. Database & Storage

- **Relational Database:** PostgreSQL (for structured data like orders, invoices, and user profiles)
- **NoSQL Database:** MongoDB (optional, for managing flexible product catalog data)
- **Caching:** Redis for session management and caching frequently accessed data

5.4. Third-Party Integrations

- **CRM:** HubSpot API for synchronizing customer data, sales pipeline, and support ticketing.
- **Payment Gateway:** Integration with Stripe, Razorpay, or PayPal.
- **Logistics API:** Integration with shipping providers (e.g., FedEx, Shiprocket) for real-time tracking.

5.5. Deployment & Infrastructure

- **Frontend Deployment:** Vercel, Netlify, or AWS Amplify
- **Backend Deployment:** AWS (EC2, Lambda, or Elastic Beanstalk), DigitalOcean, or a containerized approach with Kubernetes
- **Database Hosting:** AWS RDS for PostgreSQL or MongoDB Atlas
- **CI/CD:** GitHub Actions, GitLab CI, or other continuous integration tools to streamline deployment

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6. User Roles & Permissions

Role	Permissions
Admin	Full control over platform settings, wholesaler management, order processing, product catalog, pricing, and reporting.
Wholesaler	Access to the wholesaler portal, including order placement, invoice viewing, payment tracking, and order insights.
Guest	Limited to viewing the landing page and submitting requests to become a wholesaler.

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7. User Flows & Wireframes

7.1. Wholesaler Onboarding Flow

1. **Visitor Interaction:**
    - User visits the landing page and reviews business details.
    - User submits a “Become a Wholesaler” request.
2. **Admin Review:**
    - Admin receives the request in the Admin Portal.
    - Admin approves/rejects the application or sends an invite.
3. **Account Creation:**
    - Upon approval/invite, user registers using a secure sign-up process.
    - Wholesaler gains access to the dashboard.

7.2. Order Management Flow

1. **Order Placement:**
    - Wholesaler logs into the portal, selects products, and places a bulk order.
2. **Order Approval:**
    - Admin reviews the order, generates an invoice, and updates order status.
3. **Payment & Shipment:**
    - Wholesaler makes payment via an integrated gateway.
    - Shipment details and tracking information are updated in real time.
4. **Post-Order Analytics:**
    - Wholesaler can view order history, payment statuses, and performance analytics.

7.3. Admin Management Flow

1. **Wholesaler Request Handling:**
    - Admin reviews incoming requests and sends invitations as needed.
2. **Product & Pricing Management:**
    - Admin adds/updates products, categories, and sets pricing tiers.
3. **Order Processing:**
    - Admin monitors incoming orders, manages invoicing, and updates shipment details.
4. **Reporting:**
    - Access to dashboards displaying sales, inventory levels, and wholesaler performance.

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8. Data Model & Database Schema

8.1. Key Entities

- **Users:**

    - Fields: userID, name, email, role (admin/wholesaler), authentication credentials, profile details.
- **Products:**
    - Fields: productID, name, description, price, category, subcategory, inventory count, images.
- **Orders:**
    - Fields: orderID, userID (wholesaler), product details, quantities, order status, invoice details, timestamps.
- **Invoices:**
    - Fields: invoiceID, orderID, amount, payment status, issued date, due date.
- **Wholesaler Requests:**
    - Fields: requestID, user details, request status, timestamps.
- **Pricing Tiers:**
    - Fields: tierID, description, discount rules, applicable product categories.

8.2. Relationships

- **Users ↔ Orders:** One-to-many (a wholesaler can have multiple orders).
- **Orders ↔ Invoices:** One-to-one (each order has a corresponding invoice).
- **Products ↔ Categories/Subcategories:** Many-to-one (each product belongs to one category/subcategory).

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9. Third-Party Integrations

9.1. HubSpot CRM

- **Purpose:**
    - Sync customer inquiries, wholesaler data, and sales pipeline information.
- **Integration Points:**
    - Wholesaler registration and request approvals.
    - Order and invoice data synchronization.

9.2. Payment Gateway

- **Purpose:**
    - Process bulk order payments securely.
- **Integration Points:**
    - Invoice generation and payment status updates.

9.3. Logistics/Tracking APIs

- **Purpose:**

    - Update shipment statuses and provide tracking details.
- **Integration Points:**
    - Order management module in the Admin Portal.

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10. Deployment, Testing & Maintenance

10.1. Deployment Strategy

- **Staging Environment:**
    - Set up separate staging for testing features prior to production rollout.
- **Continuous Integration/Deployment:**
    - Use GitHub Actions or equivalent CI/CD pipelines.
- **Monitoring & Logging:**
    - Implement tools such as New Relic or Datadog to monitor performance.

10.2. Testing

- **Unit Testing:**
    - Develop tests for both frontend and backend modules.
- **Integration Testing:**
    - Ensure smooth interaction between services (authentication, payment, HubSpot integration).
- **User Acceptance Testing (UAT):**
    - Validate functionality with a sample group of wholesalers and admins.

10.3. Maintenance & Support

- **Documentation:**
    - Comprehensive documentation for code, APIs, and deployment procedures.
- **Technical Support:**
    - Establish protocols for issue tracking and resolution.
- **Regular Updates:**
    - Scheduled reviews and updates to incorporate new features and security patches.

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11. Project Timeline & Milestones

Milestone	Duration	Key Deliverables

Planning & Requirement Gathering	2–3 weeks	Finalized requirements document, wireframes
Design Phase	3–4 weeks	UI/UX designs for landing, wholesaler, and admin portals
Development – Frontend	6–8 weeks	React/Next.js codebase, responsive UI components
Development – Backend	6–8 weeks	Node.js/TypeScript API, authentication, database integration
Integration & Testing	4–6 weeks	End-to-end testing, HubSpot, payment gateway, and logistics integration
Deployment & UAT	2–3 weeks	Staging deployment, user testing feedback
Launch & Post-Launch Support	Ongoing	Production rollout, maintenance and iterative updates

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12. Risks & Mitigation

- **Integration Complexities:**
  - *Risk:* HubSpot and payment gateway integration may encounter issues.
  - *Mitigation:* Conduct early integration testing and maintain clear documentation with third-party support channels.
- **Scalability Concerns:**
  - *Risk:* Traffic spikes or large data volumes affecting performance.
  - *Mitigation:* Use cloud services with auto-scaling and efficient caching strategies.
- **Security Vulnerabilities:**
  - *Risk:* Potential breaches affecting sensitive wholesaler and order data.
  - *Mitigation:* Implement strong encryption, regular security audits, and adhere to best practices for web security.

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13. Appendices & Glossary

13.1. Glossary

- **JWT:** JSON Web Tokens used for secure authentication.
- **RBAC:** Role-Based Access Control.

- **CRM:** Customer Relationship Management.
- **UAT:** User Acceptance Testing.

13.2. Supporting Documents

- Wireframe sketches and detailed UI mockups.
- API specifications and data flow diagrams.
- Integration manuals for HubSpot, payment gateways, and logistics providers.

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14. Conclusion

This detailed document outlines the full scope, functionalities, technical architecture, and project management strategy for the Wholesale Cutlery Distribution Platform. It ensures that all stakeholders—from developers to business managers—have a clear and actionable roadmap for building a scalable, secure, and user-friendly system that meets modern business requirements.

This document has been compiled using best practices and references from industry-standard project management resources to ensure completeness and clarity.

By submitting this document to Replit or your development team, you provide a solid foundation for building a robust wholesale distribution platform using React, Node.js, and TypeScript, with integrated HubSpot CRM support and scalable cloud architecture.